

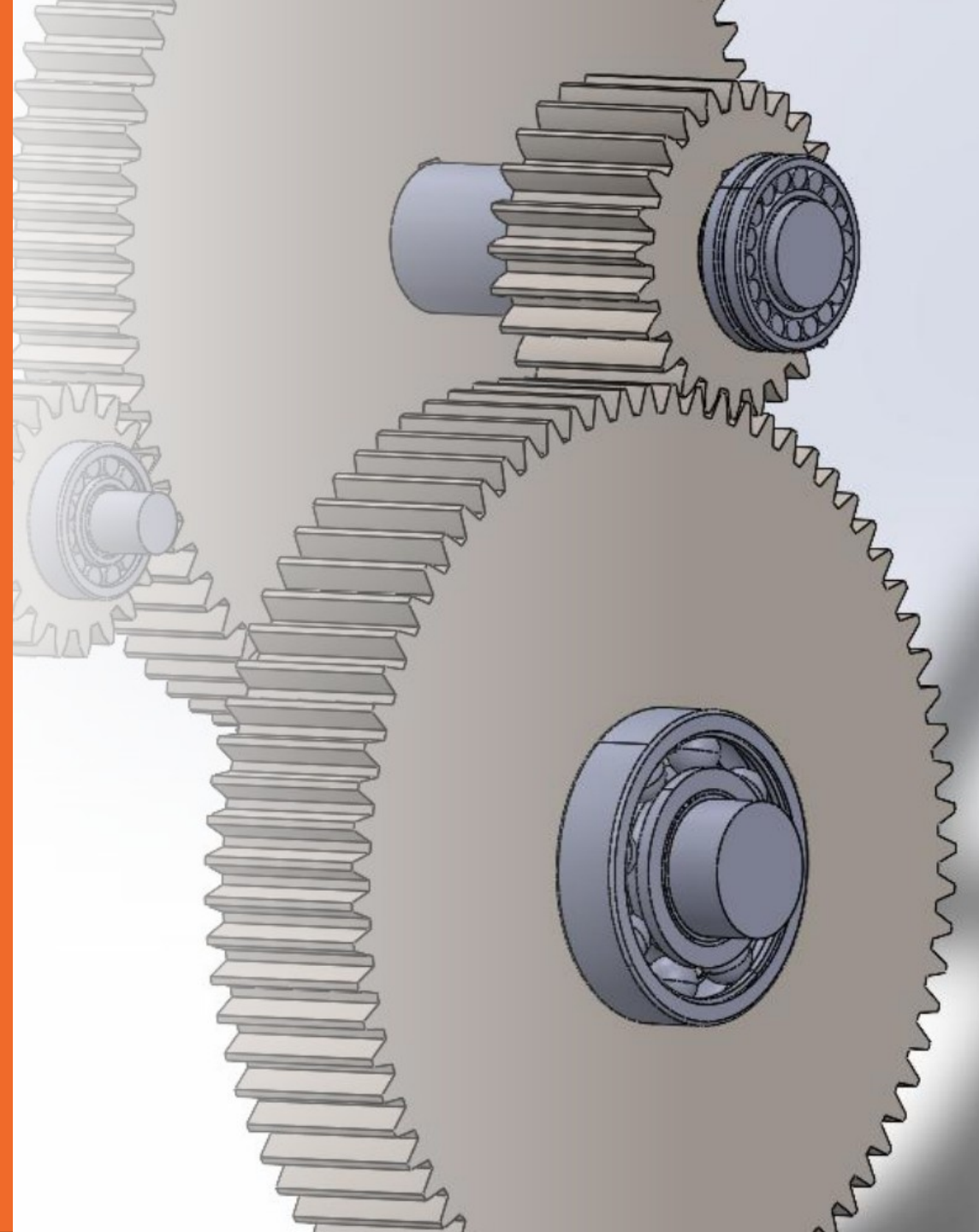
MECHANICAL ENGINEERING PORTFOLIO · 2026

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Design · Analysis · Manufacturing ·
Mechanical Systems

B.S. Mechanical Engineering
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Expected Fall 2027

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Two-Stage Spur-Gear Speed Reducer

Machine Element Design · Independent design project

WHAT

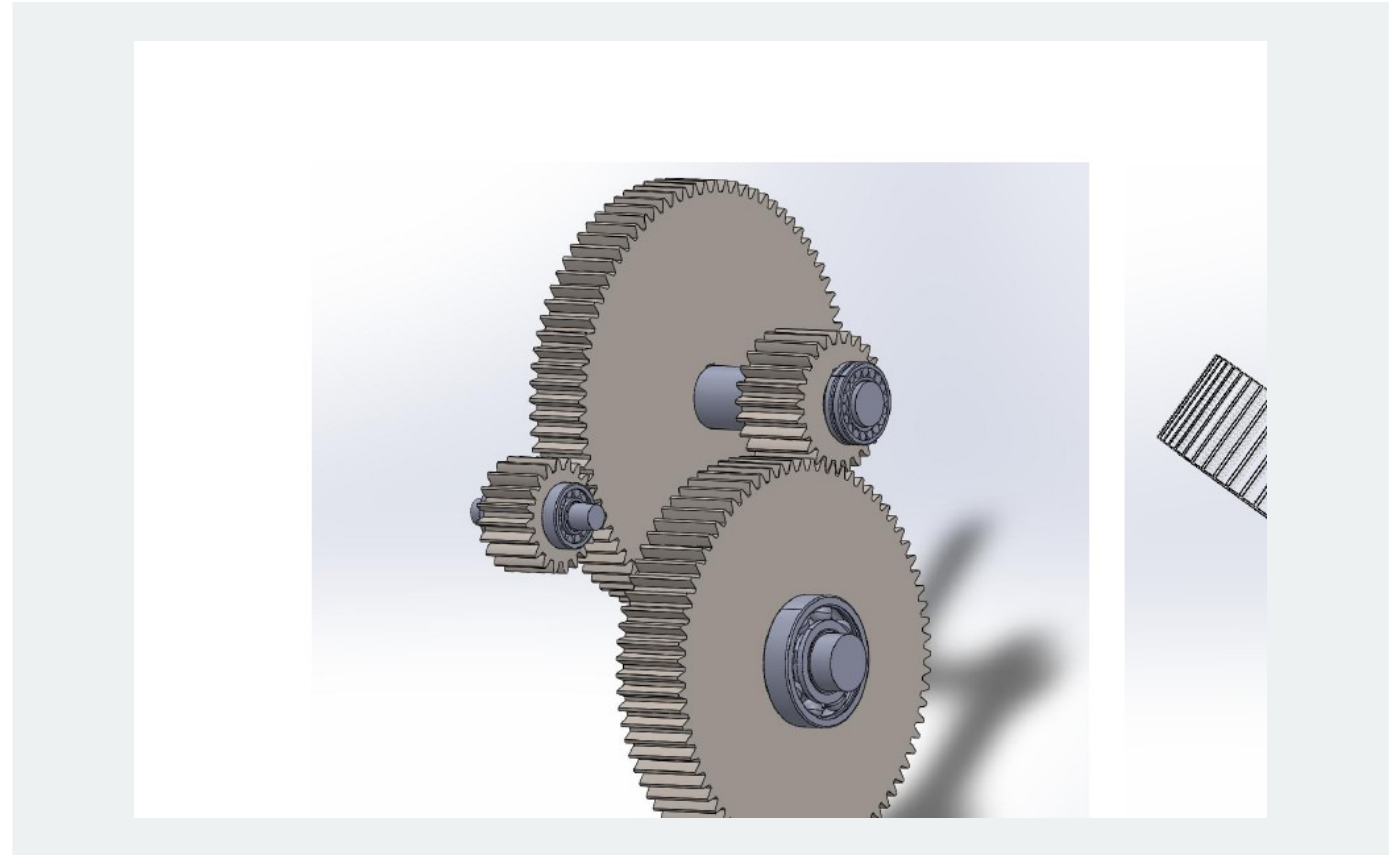
Designed an 82 hp reducer that converts 1,200 rpm input to 100 rpm output through 4:1 and 3:1 stages.

HOW

- SolidWorks assembly + shaft drawings
- AGMA bending and contact analysis
- V-M-T loading, fatigue, bearings + keys
- MATLAB beam-element FEM

RESULT

A complete concept with gears, shafts, bearings and keys sized to meet the design targets.



82 hp
INPUT POWER

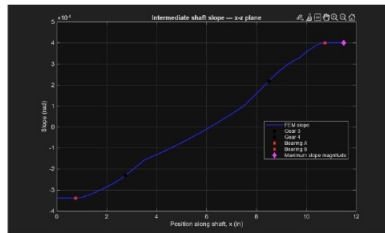
12:1
OVERALL RATIO

1,200 → 100
RPM SPEED REDUCTION

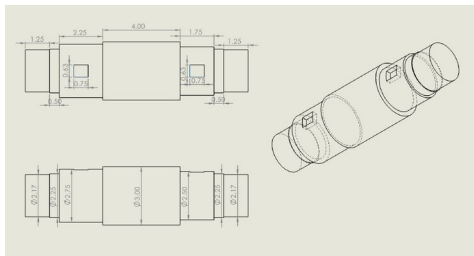
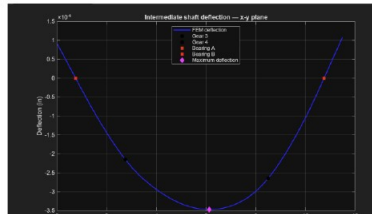
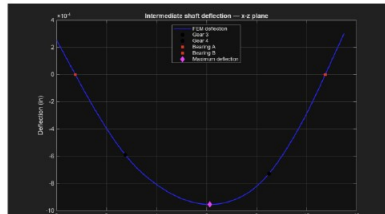
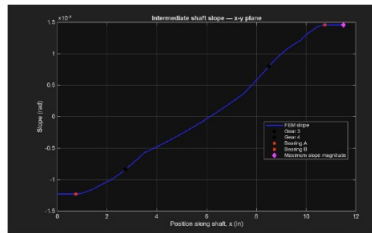
Intermediate Shaft: Fatigue + Deflection

Load cases and analytical sizing connected directly to CAD decisions

I.M Shaft x-z plane



I.M Shaft x-y plane



ENGINEERING CHECKS

Gear contact FOS

1.24

critical gear · ≥ 1.20 required

Minimum shaft FOS

≥ 1.37

Maximum shaft deflection

< 0.001 in

slope $< 4.3 \times 10^{-4}$ rad

Selected NJ307, 22211E and 6315 bearings; shaft diameters: 1.50, 2.75 and 3.00 in.

Arduino DC Motor Speed Control

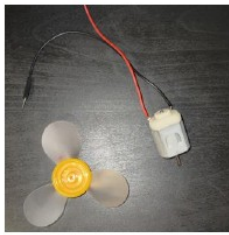
Team laboratory project · PWM control, sensing and experimental validation

BUILD

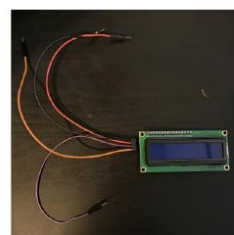
A 10 kΩ potentiometer provided the input. An Arduino Uno converted it to PWM, driving a DC motor through a 2N2222 transistor and flyback-diode circuit while an LCD displayed system values.



ARDUINO



MOTOR + FAN



LCD OUTPUT

TEST + RESULT

Four trials per operating point showed a strong voltage-speed relationship. The low-voltage response exposed the 2N2222 switching limit and motivated an L293D driver recommendation.

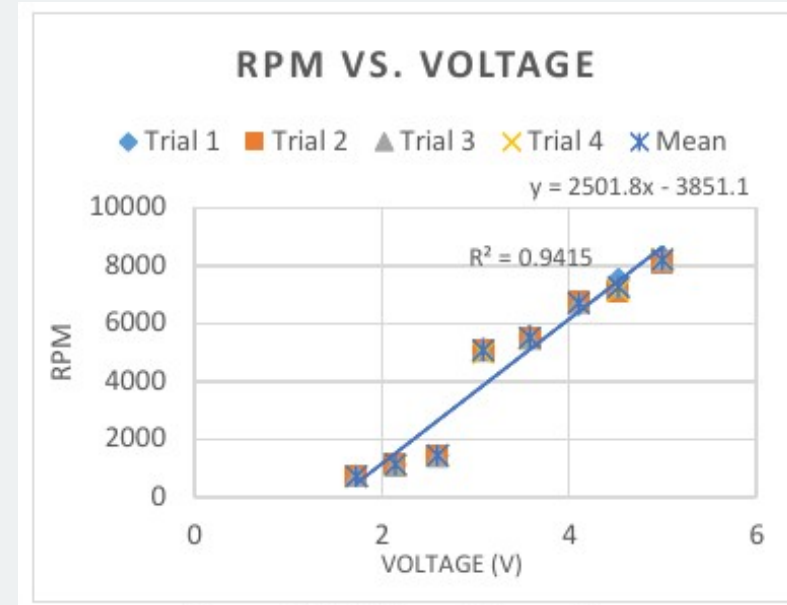


Figure 8: DT-830B and DT-2234C+

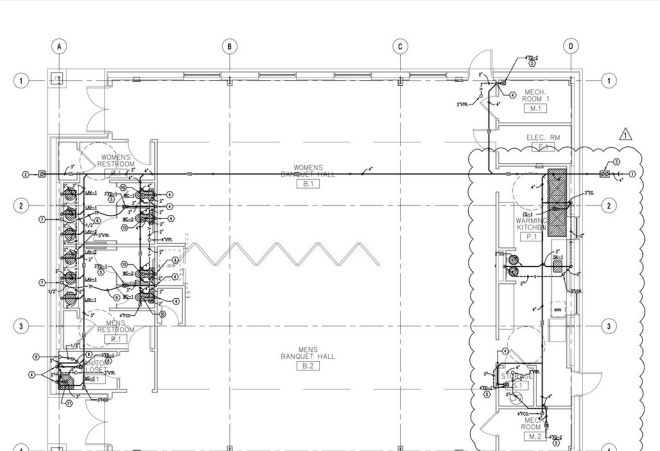
743.6 → 8,225 rpm

mean speed range

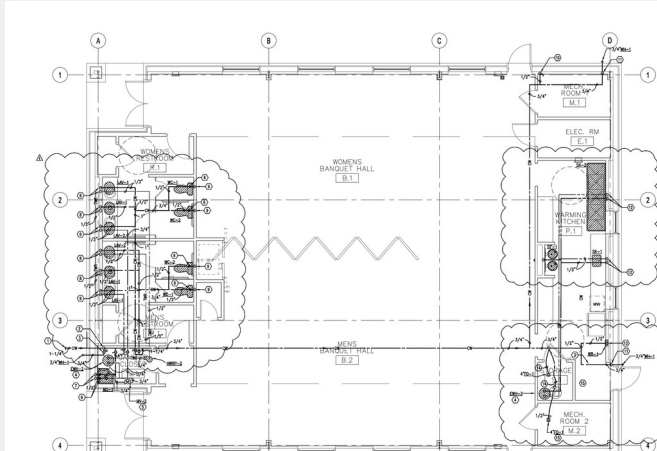
$R^2 = 0.9415$

Masjid Al-Ekhlās Permit Revision

Original Delta 1 design work · Approved for portfolio use



P.3 · SANITARY PLAN



P.5 · DOMESTIC WATER PLAN

DESIGN TRIGGER

Architectural floor-plan changes required new plumbing fixtures, relocated equipment, revised routing and removal of obsolete work.

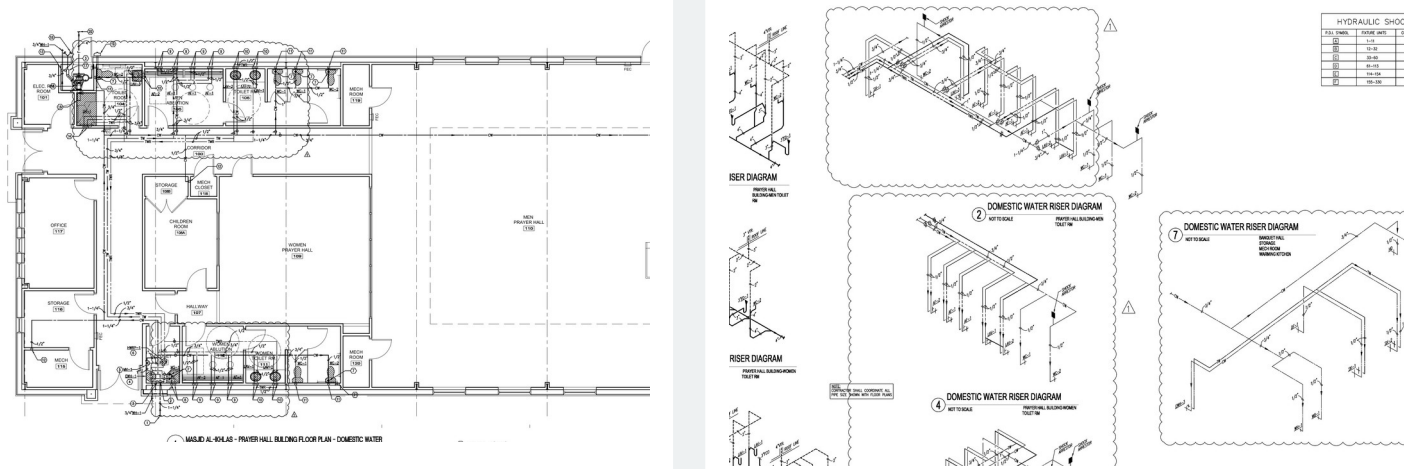
MY DELTA 1 SCOPE

- Added DF-1, SK-1, SK-2 and FS-1 with new sanitary, vent and domestic-water connections
- Relocated EWH-3 and a floor drain; added HWRP-2 and tempered-water-return circulation
- Removed redundant hot-water and obsolete fire-protection work

Every revision cloud tagged $\Delta 1$ identifies my design and documentation work.

Plans, Risers, Details + Schedules

Design intent carried across the P.3-P.9 permit-set sheets



P.4 • RECIRCULATION ROUTING

P.8 • RISER DIAGRAMS

P.9 · SCHEDULE COORDINATION

COORDINATED CHANGES

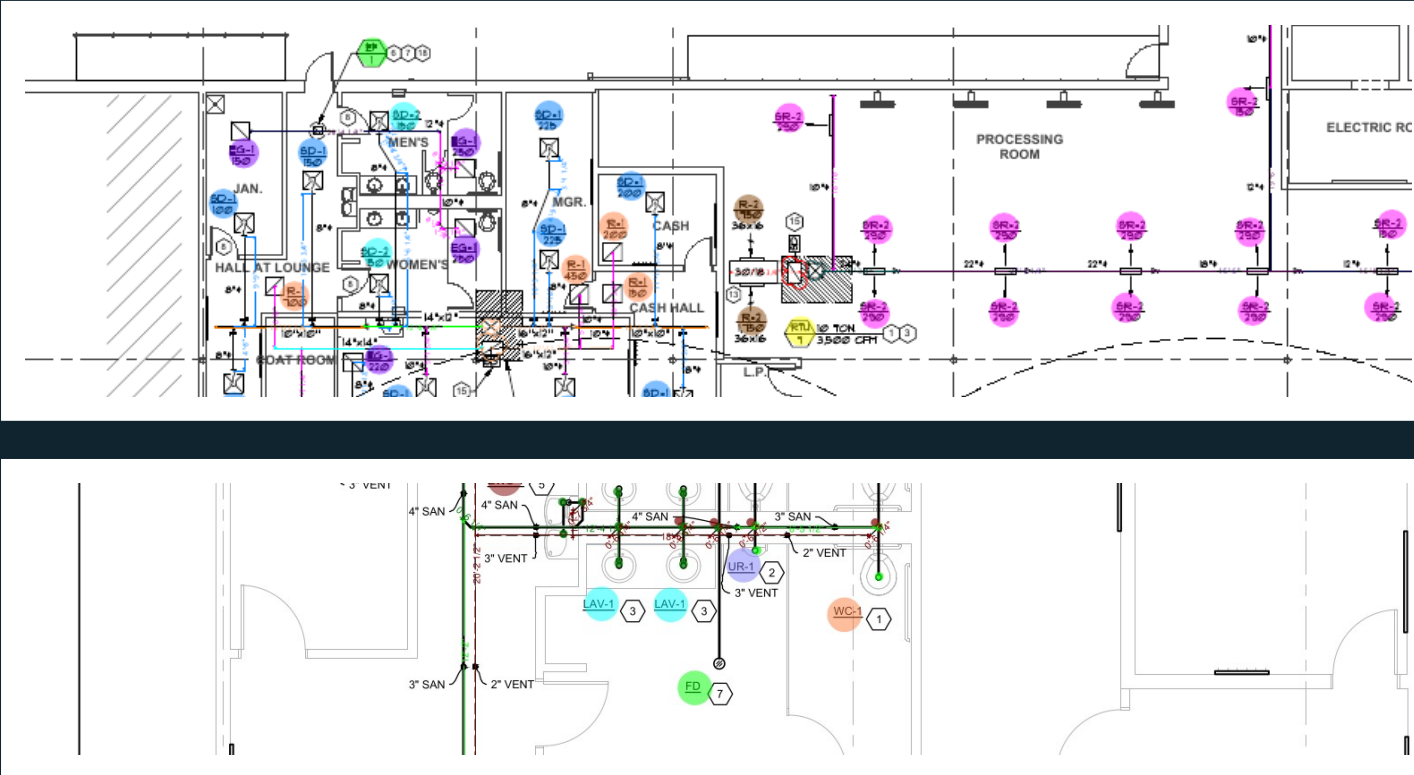
- Extended ½-in. TWR loops through men's and women's lavatories to maintain circulation
- Updated risers and created a new domestic-water riser for the storage, mechanical and warming-kitchen areas
- Added a water-heater detail, valves, line tags and permit-response notes
- Updated water-heater, recirculating-pump, fixture and sanitary-load schedules

RESULT

A coordinated revision package spanning plans, details, risers and schedules—not an isolated markup.

Commercial MEP Markup + Preconstruction

Bluebeam review, quantity takeoffs, revision tracking and equipment verification



Representative markup excerpts; identifying title blocks and project names removed.

CONTRIBUTION

- Reviewed commercial HVAC/plumbing bid drawings in Bluebeam
- Completed quantity takeoffs for ductwork, piping, equipment and fixtures
- Tracked revisions, scope changes and bid updates
- Supported plumbing plan and schedule revisions from redlines under PE review

ADDITIONAL HANDS-ON WORK

1994 Ford F-150 restoration — fuel, ignition, starting, electrical and mechanical diagnostics.

PRIVATE DRAFT · Confirm employer approval before public distribution

Shell-and-Tube Heat Exchanger CAD Study

Tutorial-guided model · Independently corrected flow separation

SCOPE

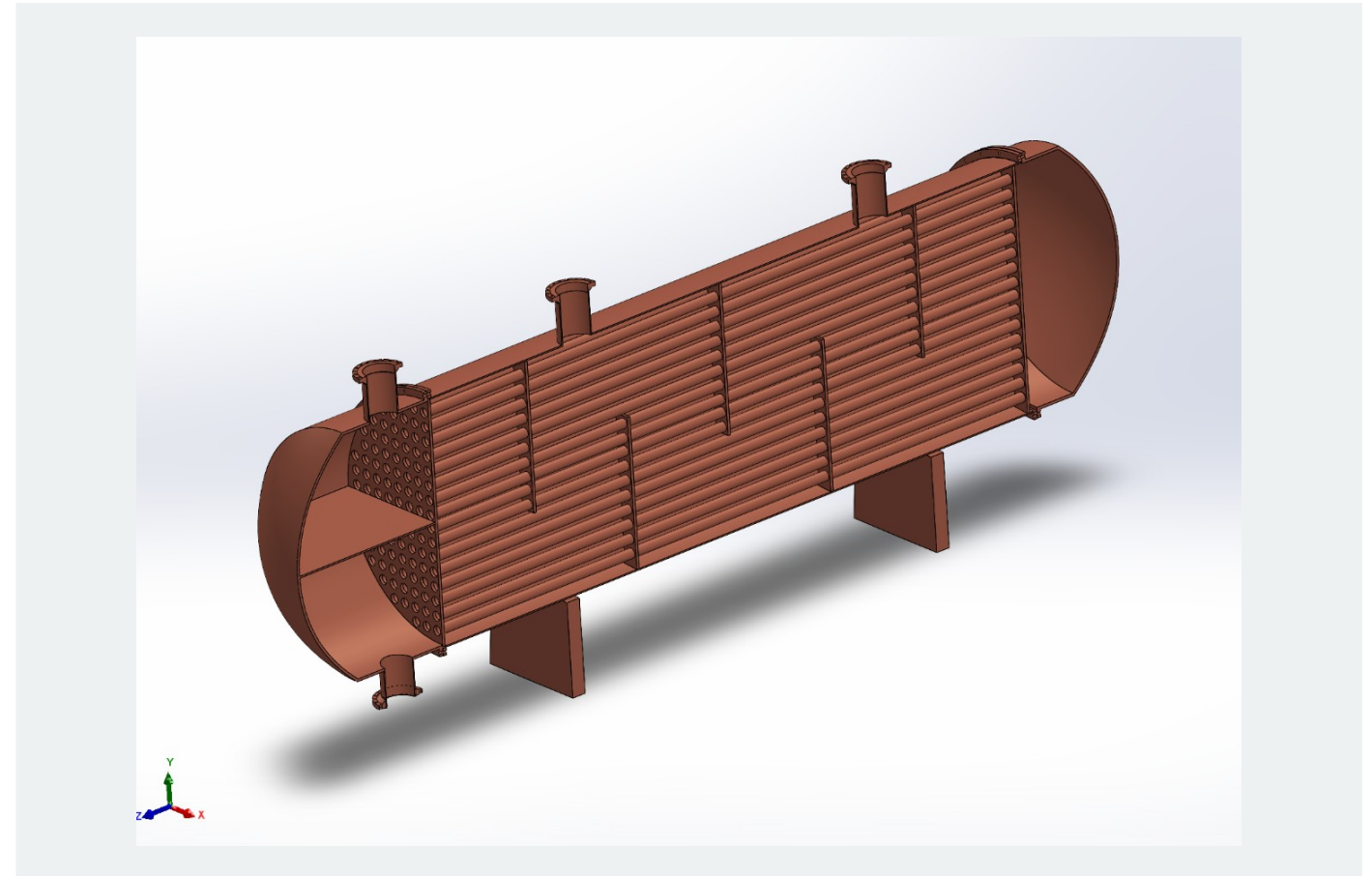
Recreated a multi-component shell-and-tube heat exchanger in SolidWorks as a guided CAD study.

INDEPENDENT CORRECTION

Identified that the tutorial configuration did not properly separate the internal flow regions, then revised the assembly so the shell-side and tube-side paths remained distinct.

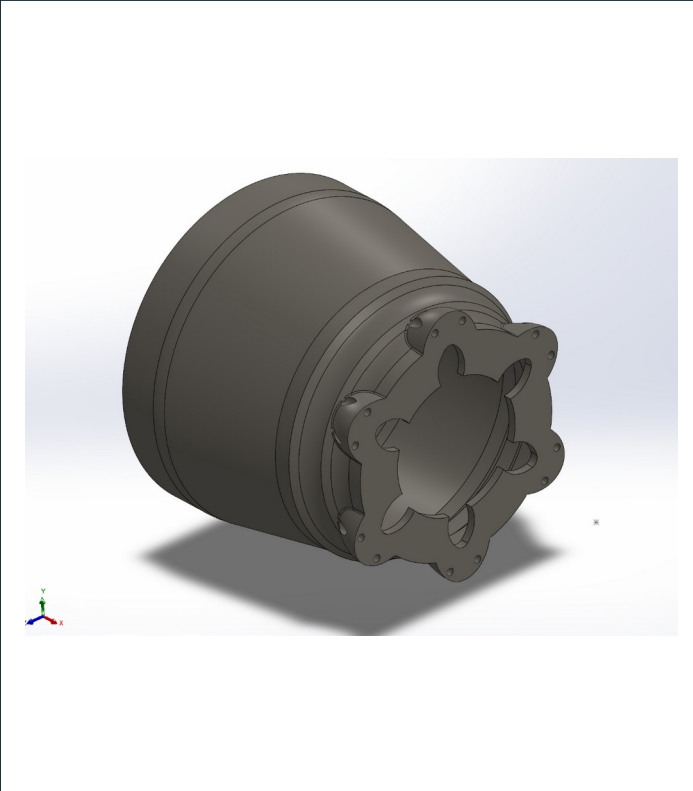
SKILLS DEMONSTRATED

Multi-part modeling · assembly mates · repeated geometry · section review · troubleshooting design intent

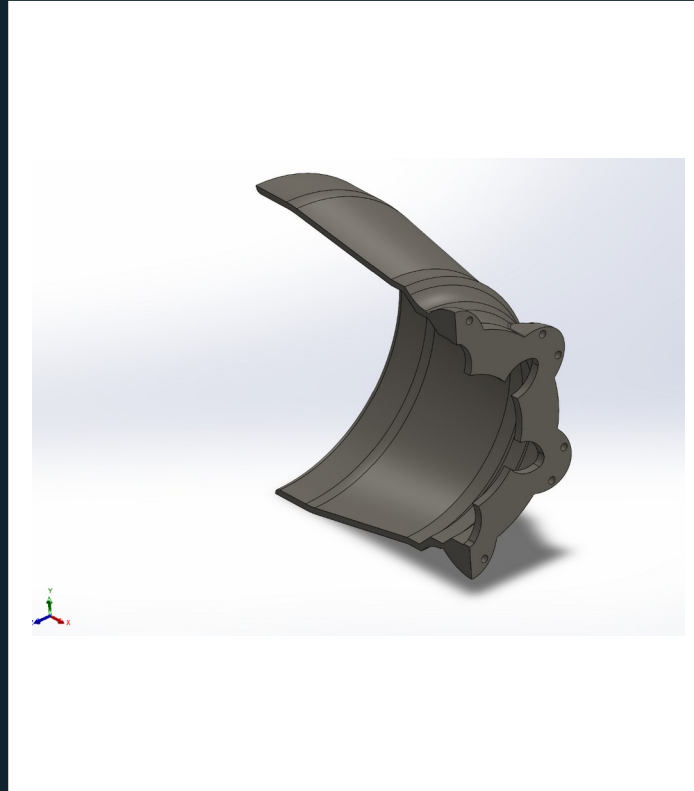


Jet Engine Rear Casing

Work in progress · Single-component SolidWorks modeling exercise



EXTERIOR



SECTION VIEW

CURRENT SCOPE

Modeled the rear casing only as practice in translating complex rotational and patterned geometry into a manufacturable-looking solid.

FEATURES USED

Revolve · boss-extrude · sweep · cut-extrude · fillets · circular pattern · face edits

STATUS

Rear-casing design practice—not presented as a completed jet-engine assembly.